

The party system effects of unstable electoral rules in Latin America

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Abstract

Prior studies have shown that electoral rules fail to determine the number of parties in new democracies due to the uncertainty of voters and politicians about which parties are viable. Latin American democracies, where electoral volatility has generally remained high and party institutionalization low, have been a primary example of this phenomenon. We argue that even in shifting political environments the fragmentation of the party system is likely to become consistent with the permissiveness of electoral rules and the level of social diversity as voters and party leaders accumulate experience with elections and the rate of electoral regime change declines. We find support for these propositions analyzing an original dataset on Latin American democratic elections. This article contributes to research on electoral systems and political parties by identifying the conditions under which equilibrium electoral outcomes can be expected after a transition to democracy in unstable institutional settings.

Keywords

electoral reform, electoral systems, Latin America, new democracies, party systems

Electoral studies have shown that key electoral rules, such as district magnitude and assembly size, affect party system fragmentation both independently and in interaction with extra-institutional factors. Yet evidence of these effects is weak or absent among new democracies. Latin American democratic regimes have been a prominent example of the failure of electoral systems to have their predicted effects. In a context of poor economic performance and frequent political and economic shocks, electoral volatility has generally remained high and party institutionalization low, thus affecting the predictability of electoral outcomes.

We argue that even in shifting political environments the fragmentation of the party system is likely to become consistent with the permissiveness of electoral rules and the level of social diversity as voters and party leaders accumulate experience with elections and the rate of electoral regime change declines. Whereas repeated elections provide information about the positions and survival capacity of parties, the maintenance of basic electoral rules improve learning their effects among voters and party leaders. We find support for these propositions analyzing an original dataset on Latin American democratic elections from the start of the latest cycle of democratization in each country of

the region up to 2021. This article contributes to research on electoral systems and political parties by identifying the conditions under which equilibrium electoral outcomes can be expected after a transition to democracy in unstable institutional settings.

We start by discussing the current research agenda on the party system effects of electoral rules in new democracies and articulating hypotheses about the impact of electoral experience and electoral regime stability on electoral outcomes. In the second section, we present a new dataset containing 198 democratic elections and 42 electoral regimes in 18 Latin American countries. In the third and fourth section we consider the impact of experience with elections and the rate of electoral regime change on the effect that electoral rules and social

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Data Availability Statement included at the end of the article

cleavages have on the number of electoral parties. In the final section, we conclude by discussing the implications of our findings for understanding electoral system effects in new democracies.

Explaining the number of parties in new democracies

While some scholars have emphasized social cleavages as the main variables that determine the number of parties in a democratic regime, most students of electoral systems have focused on key electoral rules to account for or even predict party system configurations. A perspective that became dominant over time sees the number of parties as the product of both electoral rules and social cleavages. However, whether institutions, social cleavages or their interaction is emphasized, it is unclear if theoretical expectations should be the same for old and new democracies.

The institutionalist agenda was initially advanced by Duverger, who proposed his so-called “law” that a single-member plurality (SMP) system tends toward party dualism. He also hypothesized that two-ballot majority and proportional representation (PR) systems tend toward multipartyism (Duverger, 1954: 232-233). In his view, electoral institutions shape party systems through a mechanical and a psychological effect (Duverger, 1954: 224). While the psychological effect influences party and voter strategies in anticipation of the fact that votes for minor parties might not be translated into seats, the mechanical effect constrains the seats that can be awarded from distributions of votes. Because the mechanical effect is stronger in plurality than in PR systems, the former has a discernible psychological impact: voters tend to cast a vote for their second-order preferences to avoid wasting their votes and political leaders tend not to launch an independent party into the competition to avoid losing an election against stronger competitors (Cox, 1997; Duverger, 1954).

Works that followed Duverger singled out the district magnitude rather than the electoral formula as the most important institutional determinant of the number of parties and proposed that the larger the number of seats contested in a district, the greater the number of parties that could win votes and seats in that district (Rae, 1967; Taagepera and Shugart, 1989). More recently, Taagepera (2007) proposed the “seat product” or the product of the average national district magnitude and the assembly size as the key quantity from which it is possible to logically predict the average number of seat-winning parties in simple single-tier electoral systems. Li and Shugart (2016) and Shugart and Taagepera (2017) refined the seat product model to derive from it not only the number of parties that win seats but also the number of vote-earning parties in both simple and complex electoral systems.

The expected impact of electoral institutions on the number of parties has been well documented in classic works analyzing established democracies. Lijphart (1990, 1994), for instance, shows that among these democracies PR increases the number of electoral parties compared to plurality or majority formulas and that there is a positive relationship between district magnitude and multipartyism. It is contested, however, whether these findings apply to new democracies.

Teorell and Lindstedt (2010) find that the extent to which district magnitude affects the correspondence between the number of parties in the electorate and in the legislature (i.e., the mechanical effect) is only robust to different measurements of district magnitude when we exclude from the sample countries that did not become democracies until the 1990s. Using evidence from post-communist states, Moser (1999) shows that plurality and majoritarian electoral systems fail to have a reductive effect on the number of electoral and parliamentary parties in new democracies with weak levels of party institutionalization. Similarly, Singer and Stephenson (2009) demonstrate, based on a larger sample and using district-level data, that whereas small district magnitudes constrain party competition in established democracies, they have no such effect in new democracies.¹

A perspective that gradually became conventional in electoral studies combines sociological and institutional factors in the explanation of the number of vote-earning parties. Early works (Ordeshook and Shvetsova, 1994; Powell, 1982) showed that both social cleavages and institutional factors are necessary to understand why parties emerge and fade away. Yet the generalization of the analysis was limited because it focused only on district magnitude as an institutional variable and used a limited number of elections in stable European parliamentary democracies.

The interactive specification gained wider acceptance after the work by Neto and Cox (1997) and Cox (1997), who expanded the number of explanatory institutional variables to include two-tier electoral systems and presidential regimes. They showed that in a wide variety of contexts a polity will have relatively few parties only if the electoral system is not permissive or, if it is permissive, the number of social cleavages is low (Neto and Cox, 1997: 155). They also showed that proximate presidential elections have a reductive effect on the number of electoral parties only when the number of candidates competing in presidential elections is sufficiently low (Cox, 1997; Neto and Cox, 1997).

The analysis was generalized by Clark and Golder (2006: 700), who found, using a larger sample, that the marginal effect of social heterogeneity is positive and significant once we move beyond SMP systems, thus vindicating Duverger’s law. In a parallel analysis on

presidential elections, [Golder \(2006: 45\)](#) found that temporally proximate presidential elections have a reductive effect on the effective number of electoral parties when the effective number of presidential candidates is lower than three.

Despite its widespread use, however, it is unclear if the interactive model can explain party system fragmentation in new democracies. [Clark and Golder \(2006: 698\)](#) report that when only elections during the 1990s for both established and new democracies (inaugurated after 1989) are observed, the coefficient of the interaction between district magnitude and social heterogeneity has, contrary to expectations, a negative sign.² They explain this anomaly away by arguing that while the hypothesis linking electoral system permissiveness and ethnic heterogeneity to the number of parties assumes that party systems are in some sort of “equilibrium,” many of the newly democratic countries included in their models might not have reached that equilibrium yet ([Clark and Golder, 2006: 702](#)). However, they do not explore this important problem further to attempt to refine the selection of elections in new democracies.

Other studies have put into question the impact of presidential elections on the number of electoral parties. Most works that found a reductive effect of temporally proximate presidential elections on the number of electoral parties when the number of presidential candidates is low have used samples pooling elections from presidential, semi-presidential, and parliamentary democracies, treating parliamentary elections as equivalent to midterm legislative elections in regimes with direct presidential elections. However, [Elgie et al. \(2014\)](#) show that when we restrict the sample to regimes with direct presidential elections, there is no discernible restrictive effect of temporally proximate presidential elections on the number of electoral parties, regardless of the number of candidates competing in that election. Interestingly, most observations in their analysis come from new democracies.

That institutional rules by themselves or in interaction with social cleavages do not produce the same electoral outcomes in established and new democracies is not surprising. As several authors have argued, in many new democracies it is both the absence of a long-term experience with elections and poor party system institutionalization that reduces the level of reliable information that is necessary to induce strategic behavior in voters and elites (see [Carey, 2017](#); [Ferree et al., 2014](#); [Moser and Scheiner, 2012](#)). For instance, if electoral volatility is high between elections and party organizations do not last, we may observe a larger than expected number of parties competing in elections. Social diversity may still shape party system fragmentation in this scenario, but restrictive rules such as SMP will not exert a

constraining effect. This suggests that elections in new democracies might lead to an equilibrium different from what should be expected in established democracies.

Nevertheless, new democracies also evolve and mature over time so that at some point they might become more established, at least in electoral terms. Several works have found evidence that over time party leaders learn how to target voters and voters how to avoid wasting their votes, thus leading to more predictability and improved electoral coordination ([Crisp et al., 2012](#); [Riera, 2013](#); [Roussias, 2022](#); [Tavits and Annus, 2006](#)). This experience, in turn, should lead to electoral results more in line with conventional theoretical expectations (see [Raymond et al., 2016](#)). This hypothesis makes sense. But one important aspect closely interrelated with electoral experience has been neglected: the rate of electoral regime turnover.

It has been suggested that electoral institutions may themselves be off-equilibrium in new democracies because they are frequently replaced (see [Benoit and Hayden, 2004](#); [Remmer, 2008](#)). It has also been shown that the frequency of electoral change in new democracies is higher than that of established democracies ([Calvo and Negretto, 2020](#); [Colomer, 2004](#)). If this is correct, off-equilibrium electoral results in new democracies may be both cause and effect of electoral regime change. Changing electoral rules, in turn, may fail to stabilize expectations even as experience with elections accumulates and electoral institutions will not produce their expected effects, either by themselves or in interaction with social cleavages.

Experience with elections and electoral regime stability are interconnected phenomena. At the inauguration of democracy, a new electoral system is typically adopted with various levels of predictability about which parties would be viable or what patterns of electoral competition would emerge. In some cases, new democracies may face a negative feedback loop. Early rounds of elections might produce unexpected electoral results, which in turn may lead to successive cycles of electoral reform. But electoral reforms themselves contribute to producing unexpected results, thus making electoral changes increasingly frustrating and electoral results more erratic. In other cases, party systems may gradually converge toward a level of fragmentation that is consistent with existing electoral rules, so that parties have less incentives to change them. In either situation, experience with elections and electoral regime stability are likely to reinforce each other over time.

Electoral uncertainty is supposed to be high in the early years after a transition to democracy. Yet patterns may vary. If electoral and party volatility is initially low, experience with elections will provide reliable information to voters and party leaders. This will make party systems increasingly more congruent with the electoral

system, thus reinforcing the stability of the latter. But even when electoral and party volatility is initially high, a learning process might take place. In unstable environments, repeated elections may still provide voters with incremental information about the position of parties and their capacity to survive and party leaders with information about voters' preferences, thus making electoral results more predictable. This, in turn, could reduce the temptation to tinker with electoral regimes, making them more stable. Following the logic of these arguments, we propose two hypotheses:

H1: The fragmentation of the party system in new democracies is likely to become consistent with the permissiveness of electoral rules and the level of social diversity when party leaders and voters' experience with elections increases.

H2: The fragmentation of the party system in new democracies is likely to become consistent with the permissiveness of electoral rules and the level of social diversity when electoral regimes remain stable.

The new democracies of Latin America are optimal for testing these hypotheses. Several works have shown that district magnitude matters little for explaining party system fragmentation in Latin American democracies, and that other non-institutional factors such as country-specific political cleavages are more consequential (see Coppedge, 1997; Shugart and Mainwaring, 1997).³ If social heterogeneity accounts for party system fragmentation, Latin American democracies present an ideal variation, from relatively homogeneous Argentina, Chile, and Uruguay to highly diverse Bolivia, Ecuador, and Peru. However, in a literature review on electoral laws and party systems in Latin America, Morgenstern and Vázquez-D'Elia (2007) conclude that none of the typical institutional variables such as district magnitude or the concurrence of legislative and executive elections seem to explain party fragmentation in the region, even after controlling for social cleavages. Similarly, in a study on the causes and consequences of electoral reform in Latin America from 1978 to 2002, Remmer (2008) finds that whereas changes in the party system effectively explain the occurrence of electoral reforms in the region, the latter do not produce the expected effects on the number of parties; neither permissive reforms increase party fragmentation nor restrictive ones reduce it.

New Latin American democracies are also interesting for analyzing the party system effects of electoral rules because electoral volatility has generally not decreased over time in this region. It has been argued that the stabilization of party systems in new democracies is a function of time (Converse, 1969; Tavits, 2005). Yet Roberts and Wibbels (1999) showed that electoral volatility in Latin America

increased during the 1990s although several democracies in the region had at that time endured for a decade or more. More recent data by Mainwaring and Su (2021) confirms persistent electoral volatility in the region, which is determined by factors such as poor economic growth, permissive electoral systems and large number of parties, low levels of party identification, and a perception of pervasive corruption among voters. These contextual factors make finding equilibrium in electoral results challenging, even as democracies age.

Data on democratic elections and electoral regimes in Latin America

To explore the party system effects of electoral rules in new Latin American democracies, we created a novel dataset containing information on electoral regimes and election results for 198 democratic elections of the lower or single chamber of the legislature held in eighteen Latin American countries between the last inauguration of democracy (1979 or later for most countries, but a few before that date) and 2021.⁴ Elections were considered democratic when both the chief executive and the legislature were chosen by popular election, voters were offered multiple options on ballots, and the contest was fair in the sense that there was no evidence of fraud or of incumbents abusing power to prevent opponents from winning.⁵ The elections included in the database were coded by both country and date, as well as by the electoral regime under which they were held.

As regimes have various components that may undergo variations of different magnitudes over time, it is unclear when an electoral regime remains stable or experiences change. For instance, Lijphart (1994: 52) finds evidence of electoral stability in the fact that no country changed from plurality to PR or vice versa in his study of twenty-seven stable electoral democracies from 1945 to 1990. Katz (2005), however, argues that electoral system change, even among established democracies, is considerably more frequent. Just by including the electoral formula through which a strong president is elected along with the formula to elect the lower or single chamber of the legislature, he counts 14 major electoral reforms among established democracies in the world during the period 1950-2005 (2005: 58).

The age of a democracy also affects electoral regime stability. Comparative evidence indicates that the rate of electoral system change is greater among newer than among older democracies (see Colomer, 2004). Latin America is a salient example. Merely between 1978 and 2008, new democracies in Latin America underwent 45 major electoral changes in their systems to elect presidents and deputies (Negretto, 2013: 25-29). These reforms refer only to

significant alterations of the electoral formula, the adoption or elimination of legal thresholds, and a twenty-five percent or larger change in the size of district magnitudes or assembly size. Another study including additional components of electoral systems, counts 55 major electoral reform events in the rules to elect Lower House members from 1978 to 2015 (Calvo and Negretto, 2020).

We define the electoral regime to elect legislators as the set of rules that determine the size of the lower or single chamber of congress, the average national district magnitude for the election of that chamber in the lower or single tier, and the existence and size of an upper tier. Following Lijphart (1994: 13-14), we consider that an electoral regime changes when there is an alteration of at least 20% or more in the size of the lower or single chamber of congress or in the average district magnitude for the election of that chamber in the lower or single tier. We also count as reform the creation or elimination of an upper tier and an alteration of 20% or more in the number of legislators elected in that tier. This conceptualization of change is adequate to capture potential shifts in proportionality in contexts where there are no pure majoritarian systems, as is the case of Latin American democracies.⁶ It also fits the identification of electoral regimes using the two central variables proposed by Taagepera (2007), with the addition of the upper tier as a possible relevant component (Cox 1997).⁷

Based on these concepts, we identify 42 regimes used to elect deputies in the 198 democratic elections included in the dataset. Following our definition of electoral change, we count a total of 34 significant electoral reforms during the period under analysis. Table 1 summarizes the number of elections and the number of electoral regimes to elect deputies by country included in the database.

There is a huge temporal and cross-national variation in electoral system instability. Out of the 34 electoral reforms observed, 15 took place before 1990, 10 between 1990 and 2000, eight between 2000 and 2010, and one between 2010 and 2021. This trend strongly suggests that several new democracies in Latin America passed through an early period of change and experimentation and gradually entered a period of higher institutional stability. A few countries have had consistently stable or unstable electoral regimes. Whereas Argentina, Brazil, Mexico, Panama, and Uruguay maintained the same electoral regime, the Dominican Republic, Ecuador, Guatemala, and Venezuela introduced more than three reforms since the transition to democracy. The rest experienced between one and three reform events. Overall, the mean number of elections per electoral regime in Latin America during this period was 4.7, again with considerable variations across countries, from a low of just 2.3 elections per regime (Guatemala) to a high of 20 (Argentina).

Table 1. Competitive Electoral Regimes and Elections in Latin America: From the inauguration of democracy to 2021.

Country	Number of elections	Number of electoral regimes lower chamber	Mean number of elections per regime
Argentina	20	1	20
Bolivia	11	3	3.66
Brazil	9	1	9
Chile	9	2	4.5
Colombia	14	2	7
Costa Rica	18	2	9
Dom. Rep	12	5	2.4
Ecuador	15	4	3.75
El Salvador	13	3	4.33
Guatemala	9	4	2.25
Honduras	11	2	5.5
Mexico	9	1	9
Nicaragua	7	2	3.5
Panama	7	1	7
Paraguay	7	2	3.5
Peru	8	2	4
Uruguay	8	1	8
Venezuela	11	4	2.75
Total	198	42	
Average	11	2.3	4.7

Source: Authors.

Statistical analysis

To test how experience with democratic elections and electoral regime stability affects the impact of electoral rules on party systems in new Latin American democracies, we analyze our dataset in two different ways. First, we run tests comparing results using the full sample with subsamples that distinguish between early and late elections and between elections that took place under the same or different electoral regimes. Second, we use only the full sample to estimate the impact of variables that capture the accumulation of electoral experience and electoral regime stability.

Given the structure of the data, we use a pooled OLS regression with robust standard errors clustered by country. The main dependent variable is the effective number of electoral parties (ENPv), measured according to the [Laakso and Taagepera index \(1979\)](#). Although the fragmentation of the party system can be disaggregated into the effective number of vote-earning and seat-winning parties, we focus on the former because the impact of the electoral system on the parties competing in legislative elections is potentially mediated by several non-institutional and contextual variables that affect the behavior of voters and party leaders.

Following standard practice in the literature, in the models that compare results using the full sample with subsamples or early and late elections and subsamples of elections under the same or different electoral regimes, the main variables are the effective number of ethnic groups (ethnic), which is a proxy of the level of social diversity, and the average national district magnitude ($\ln(M)$), which captures the permissiveness of the electoral regime. The first is measured using an index of ethnic fragmentation proposed by [Fearon \(2003\)](#) and the second by taking the logarithm of the average district magnitude in the first or single tier. We also include the conventional interaction term between the two variables to estimate whether party fragmentation is conditional on both the permissiveness of electoral rules and the degree of social diversity. We discuss in the text and report in the online appendix the results of models that test the impact of institutions alone.

In the analysis of the impact of electoral experience using the full dataset, we add a variable (number) that captures the number of competitive elections that took place since the inauguration of democracy, regardless of whether they were regulated by the same or different electoral rules. To assess the impact of electoral regime stability using the full dataset, we include a variable (stability) that reflects the number of elections accumulated under the same rules. This variable focuses on institutional stability mainly because the counting of elections starts anew every time that an electoral regime

changes. Each of these variables is then interacted with both the effective number of ethnic groups and the log of the average national district magnitude.

The control variables include the percentage of assembly seats allocated in an upper tier (upper), the Laakso and Taagepera index ([1979](#)) of the effective number of candidates (enc) competing in the presidential election, the proportion of time that separates presidential elections and legislative elections (proximity) measured according to the formula proposed by [Neto and Cox \(1997\)](#), and two additional interaction terms: ethnic*upper and enc*proximity. We also control for the number of years elapsed since the first democratic election (time) because there is evidence that electoral systems in Latin America have become more permissive and party systems more fragmented over time (see [Coppedge, 1997](#); [Negretto, 2013](#); [Remmer, 2008](#)).⁸

Experience with democratic elections under new electoral rules

To analyze the impact of electoral experience, we start by segmenting the data between early and late elections. This selection of observations builds on previous work ([Raymond et al., 2016](#); [Roussias, 2022](#)), which has shown that it takes about five elections for voters and party leaders to learn how to behave strategically in electoral competition. We thus consider the first five elections after the transition to democracy in every country as early elections, and the rest as late ones.⁹ [Table 2](#) shows the results comparing the full sample containing all elections (198) with a subsample of early elections (90) and a subsample of late elections (108).

The first result to note is that in no case ethnic heterogeneity significantly increases the ENPv in SMP electoral systems (i.e., when $\ln(M) = 0$). It is important to bear in mind, however, that unlike samples including old democracies, in Latin America there are no pure SMP systems. The cases in which district magnitude equals 1 in our database refer to the majoritarian lower tier of mixed electoral systems.

The second important result is that in Model 2, that is, in the subsample of early elections, the interaction between district magnitude and social heterogeneity is negative, and as such, inconsistent with the conventional theory. The interactive variable is only positive and significant for late elections. [Figure 1](#) shows that for these elections (Model 3) the marginal effect of ethnic heterogeneity significantly increases the number of parties competing in legislative elections when the log of the average district magnitude is greater than 1.43, which is equivalent to a magnitude of 4.2. This result lends support to H1 and confirms that previous findings on the learning

Table 2. Explaining ENPv in Latin America by electoral experience.

Regressor	Electoral Institutions* <i>Cleavage structures</i>		
	Model 1	Model 2	Model 3
	All elections	Early elections	Late elections
Ethnic	1.416 (6.770)	7.499 (4.885)	−10.945 (8.468)
ln(M)	0.457 (1.028)	1.438* (0.815)	−1.576 (1.383)
Ethnic * ln(M)	1.114 (4.064)	−2.928 (2.578)	9.483* (5.375)
Upper tier	6.184 (4.842)	0.704 (5.224)	3.054 (6.090)
Enc	1.418*** (0.331)	1.439*** (0.201)	1.036** (0.413)
Proximity	0.792 (1.263)	−0.281 (1.153)	−0.116 (1.427)
Ethnic *upper	−7.122 (16.380)	−6.971 (16.485)	14.188 (19.601)
Enc*proximity	−0.335 (0.362)	−0.227 (0.327)	−0.0120 (0.395)
Time	0.026** (0.012)	−0.004 (0.020)	0.022 (0.016)
Constant	−2.558 (1.932)	−2.862** (1.282)	1.466 (2.682)
Obs	198	90	108
R2	0.517	0.601	0.645
R2 (adj.)	0.494	0.556	0.612

Note: Numbers in parentheses are robust standard errors clustered by country. *** $p < .01$ ** $p < .05$; * $p < .1$.

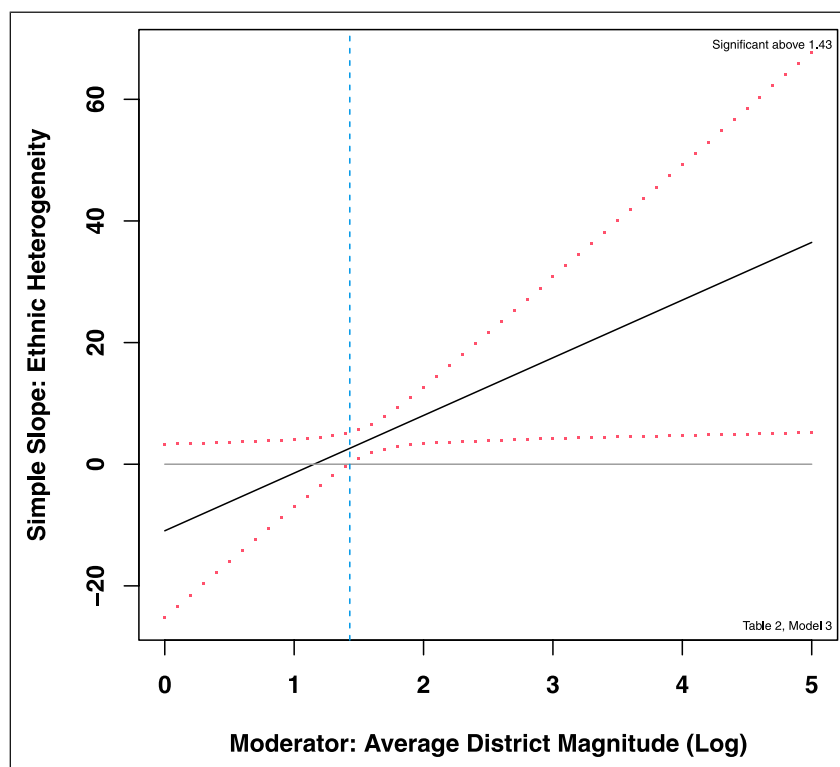


Figure 1. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude in Late Elections. Note: Red dotted lines represent a 90% confidence interval.

effects of repeated elections hold even in unstable political environments.

Even after considering experience with elections, the proximity of legislative and presidential elections does not appear to have a reductive effect on the fragmentation of electoral parties. Its interaction with the number of presidential candidates is undistinguishable from zero and has a consistently negative sign. When we plot the marginal effect of temporally proximate presidential elections on the ENPv for early and late elections (results not shown) in no case do we see that, as predicted by the theory, temporally proximate presidential elections have a reductive effect on the number of electoral parties when there are few presidential candidates.

To provide a further test of the impact of institutional rules by themselves on the number of electoral parties, we replaced the log of the average district magnitude for the log of Taagepera's seat product (2007), that is, the product of the average district magnitude and the assembly size of the single or lower chamber in the lower or single tier of elections, which arguably captures more institutional features affecting the permissiveness of the electoral system than the district magnitude alone. In these models, we drop ethnic fractionalization and replace the effective number of presidential candidates (a behavioral variable) for presidential formulas. The latter are measured on an ordinal scale from the least to the most

permissive, ranging from 1 (plurality) to 3 (majority runoff), with 2 for formulas with thresholds below absolute majority. The main finding, which we report in the online appendix (see Table A1), is that the permissiveness of the electoral system, as measured by the product of the average district magnitude and the size of the assembly, has a positive and statistically significant effect on party system fragmentation only for late elections.¹⁰ Nevertheless, the smaller proportion of the variance in the fragmentation of the number of electoral parties that is explained in this test, compared to the interactive model, suggests that non-institutional factors such as social cleavages should be taken into account.

Table 3 shows the statistical test using the full sample and adding the number of elections as a variable, which is interacted with the log of the district magnitude and the level of ethnic fragmentation. Consistent with the previous analysis, the results support our hypothesis that the fragmentation of the party system is congruent with the permissiveness of the electoral system and the diversity of the social structure, as experience with elections increases.

Figures 2 and 3 illustrate the marginal effect of ethnic fragmentation on the number of electoral parties when two elections and when nine successive elections have taken place, respectively. The effect is only positive and significant within district magnitude 2.4 and 9.2 in the latter case. Figure A1 in the online appendix shows how the marginal

Table 3. Explaining ENPv in Latin America by electoral experience.

Regressor	Electoral Institutions* Cleavage Structures* Number of elections
	All elections
Ethnic	4.943 (6.063)
ln(M)	1.489* (0.833)
Number	0.186 (0.295)
Ethnic*ln(M)	-3.010 (2.569)
Ethnic*Number	-0.997 (0.827)
ln(M)*Number	-0.212 (0.178)
Ethnic*ln(M)*Number	0.957* (0.514)
Upper tier	1.783 (5.815)
Enc	1.311*** (0.300)
Proximity	0.710 (1.195)
Ethnic*upper	5.792 (19.890)
Enc*proximity	-0.325 (0.318)
Time	0.019 (0.027)
Constant	-2.807 (1.988)
Obs	198
R2	0.585
R2 (adj.)	0.556

Note: Numbers in parentheses are robust standard errors clustered by country. *** $p < .01$; ** $p < .05$; * $p < .1$.

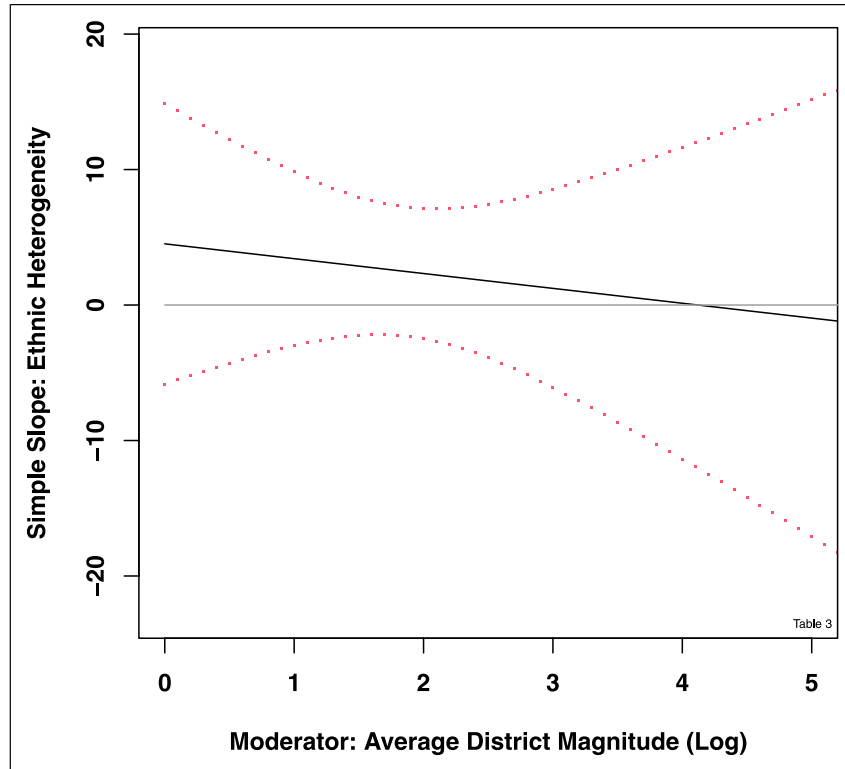


Figure 2. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude when Number = 2. Note: Red dotted lines represent a 90% confidence interval.

effect of ethnic heterogeneity on party system fragmentation at different levels of district magnitude varies as the value of the second moderator (number of elections) increases between one and twelve.

Because electoral reforms have been more frequent during early elections, the later an election takes place after the inauguration of democracy, the more likely is that this election occurs under a stable electoral regime. However, as electoral regimes may still change several years after the transition to democracy, we need a more refined measure to capture the impact of electoral system stability. We undertake this task in the next section.

Electoral regime stability

For the analysis of the impact of electoral regime stability, we start by segmenting the data depending on whether elections take place under the same or different electoral rules. Although it is unclear how many elections should be

held to observe the “equilibrium” number of parties under a given electoral regime, it seems logical to discard one-election regimes (see [Ordeshook and Shvetsova 1994: 102](#)). Yet parties and voters probably need more than one election to learn the properties of an electoral regime. [Taagepera \(1998\)](#) has suggested that at least three elections under the same rules are required to adjust behavior and expectations, a reasonable criterion that we follow. We thus make a further selection of elections based on whether three or more than three elections have been held under the same electoral regime.

[Table 4](#) shows the results comparing the full sample of all elections that took place under electoral regimes that survived two consecutive elections (123) with a subsample of early elections (37) and a subsample of late elections (86) that satisfy the same criterion. Because this selection is more demanding, the number of observations drops sharply, particularly in the case of early elections, which is consistent with the fact that most electoral reforms in Latin America

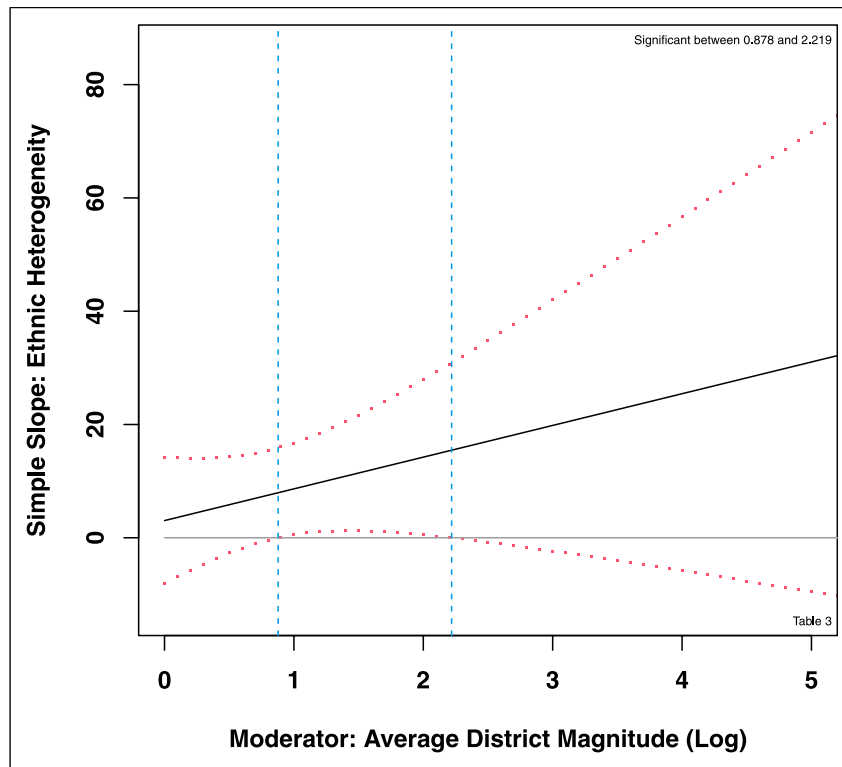


Figure 3. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude when Number = 9. Note: Red dotted lines represent a 90% confidence interval.

Table 4. Explaining ENPv by electoral experience and electoral regime stability.

Regressor	Electoral Institutions* <i>Cleavage structures</i>		
	Model 1	Model 2	Model 3
	All elections	Early elections	Late elections
Ethnic fragmentation	1.505 (7.682)	7.277 (6.031)	−8.894 (8.248)
ln(M)	0.765 (1.329)	1.863 (1.581)	−1.325 (1.295)
Ethnic fragmentation* ln(M)	2.120 (4.741)	−2.262 (3.572)	9.439* (5.026)
Upper tier	4.309 (5.728)	−0.988 (8.063)	−0.728 (6.778)
Enc	1.074*** (0.281)	1.355*** (0.437)	0.563*** (0.194)
Proximity	−0.541 (0.922)	2.333 (2.799)	−2.200** (0.800)
Ethnic fragmentation *upper	−1.719 (18.240)	−0.716 (20.170)	18.104 (19.539)
Enc*proximity	−0.078 (0.296)	−0.857 (0.682)	0.427* (0.222)
Time	0.018 (0.015)	−0.116* (0.061)	0.016 (0.017)
Constant	−1.863 (2.328)	−2.504 (1.888)	2.729 (2.169)
Obs	123	37	86
R2	0.530	0.596	0.674
R2 (adj.)	0.492	0.462	0.636

Note: Numbers in parentheses are robust standard errors clustered by country. *** $p < .01$; ** $p < .05$; * $p < .1$.

took place during the early years after the transition to democracy.

The main finding is that late elections that take place under electoral regimes that survived at least three elections produce results in line with the expected effects of the interaction between electoral rules and social diversity. Whereas in Model 2 the interaction term comes with a negative sign for early elections, Model 3 shows that the interaction between ethnic heterogeneity and district magnitude is not only positive but also statistically significant for late elections under stable electoral regimes. This is consistent with our H2.

Figure 4 below illustrates that in Model 3 the marginal effect of ethnic heterogeneity is statistically associated with a significant increase in the effective number of electoral parties as the electoral system becomes more permissive; specifically, when the log of the average district magnitude is greater than 1.35, which is equivalent to a magnitude of 3.9. Below this magnitude we find relatively restrictive electoral regimes in the sample, such as those of Bolivia since

1997, Mexico since 1994, and Venezuela from 1993 to 2005, where the deputies are elected by SMP in the basic tier; Panama since 1989, which includes many single-member districts; and the binomial electoral system used in Chile from 1989 to 2013.

As for the impact of proximate presidential elections on the effective number of electoral parties conditional on the effective number of presidential candidates, Model 3, which excludes both early elections and elections regulated by electoral regimes that did not last more than two electoral cycles, also shows the expected results. Whereas the coefficient of proximity is negative, the coefficient of the interactive term is positive, and in both cases statistically significant, as predicted by the theory. Figure 5 illustrates how the marginal effect of temporally proximate presidential elections on the effective number of electoral parties significantly reduces the effective number of electoral parties when the effective number of candidates is lower than 3.6.

We also run a test of the impact of electoral regime stability using institutional variables only. We find, as

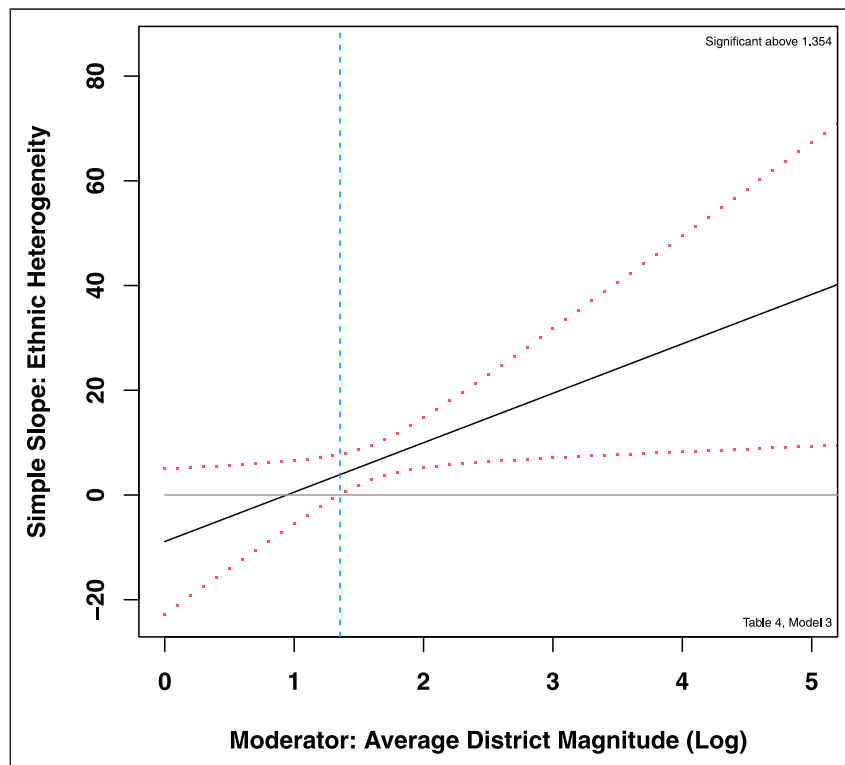


Figure 4. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude in Late Elections under the same Electoral Regime. Note: Red dotted lines represent a 90% confidence interval.

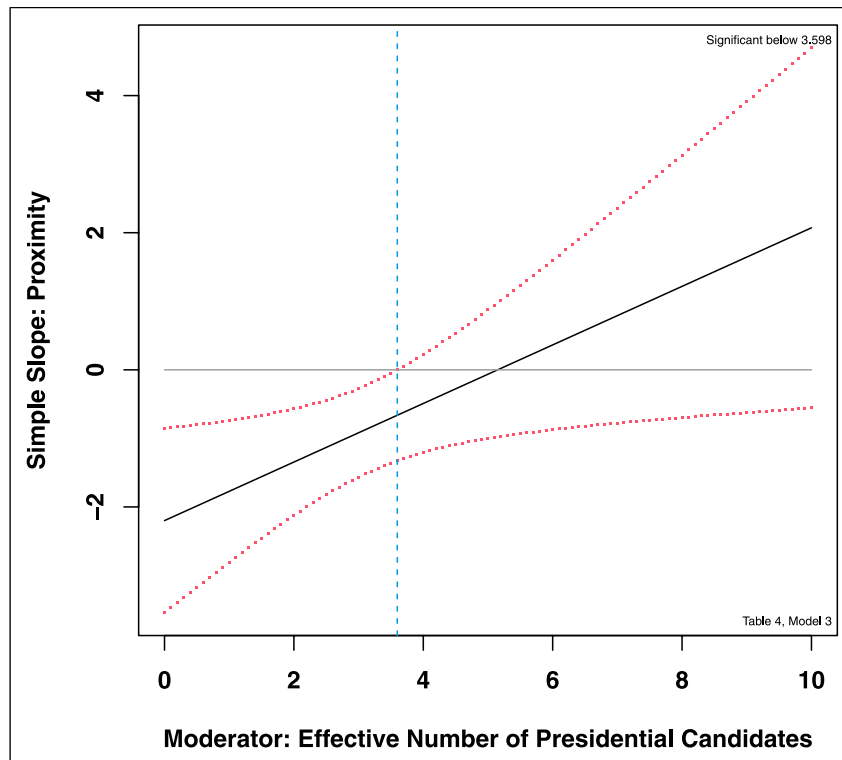


Figure 5. Marginal Effect of Temporal Proximity on the Effective Number of Electoral Parties by the Effective Number of Presidential Candidates in Late Elections under the same Electoral Regime. Note: Red dotted lines represent a 90% confidence interval.

shown in the online appendix (see Table A3), that more permissive electoral systems have a positive and significant impact on party system fragmentation only in late elections under stable electoral regimes. Similarly, the test also suggests that a model including social cleavages provides a more complete specification.

Table 5 shows the statistical test using the full sample and adding the accumulated number of elections under the same electoral regime as a variable, which is in turn interacted with the log of the district magnitude and the level of ethnic fragmentation. The results provide a strong support for our H2. The main finding is that the fragmentation of the party system becomes consistent with the permissiveness of the electoral system and the diversity of the social structure, not simply when the number of elections increases but when successive elections are held under the same electoral

regime. The size of the coefficient of the interaction between social cleavages, district magnitude, and stability is higher and more significant than when we considered just the absolute number of consecutive elections.

Figures 6 and 7 show the marginal effect of ethnic fragmentation on the number of electoral parties when two and when nine consecutive elections have taken place under the same electoral regime, respectively. The effect is only positive and significant in the latter case. Figure A2 in the online appendix shows how the marginal effect of ethnic heterogeneity on party fragmentation at different levels of district magnitude varies as the value of the second moderator (stability) increases between one and twelve. We find evidence that after the accumulation of five elections under the same set of electoral rules, the fragmentation of the party system gradually becomes more

Table 5. Explaining ENPv in Latin America by electoral regime stability.

Regressor	Electoral Institutions* All elections
	Cleavage Structures* Regime stability
Ethnic	3.658 (4.452)
ln(M)	0.834* (0.414)
Stability	-0.338 (0.280)
Ethnic * ln(M)	-3.403* (1.802)
Ethnic * stability	-0.038 (0.704)
ln(M) * stability	-0.068 (0.158)
Ethnic* ln(M)*Stability	1.090** (0.475)
Upper tier	6.327 (3.927)
Enc	1.098*** (0.248)
Proximity	-0.505 (0.784)
Ethnic * upper	-9.638 (11.300)
Enc*proximity	0.017 (0.224)
Time	0.004 (0.013)
Constant	-0.260 (1.420)
Obs	198
R2	0.685
R2 (adj.)	0.663

Note: Numbers in parentheses are robust standard errors clustered by country. *** $p < .01$; ** $p < .05$; * $p < .1$.

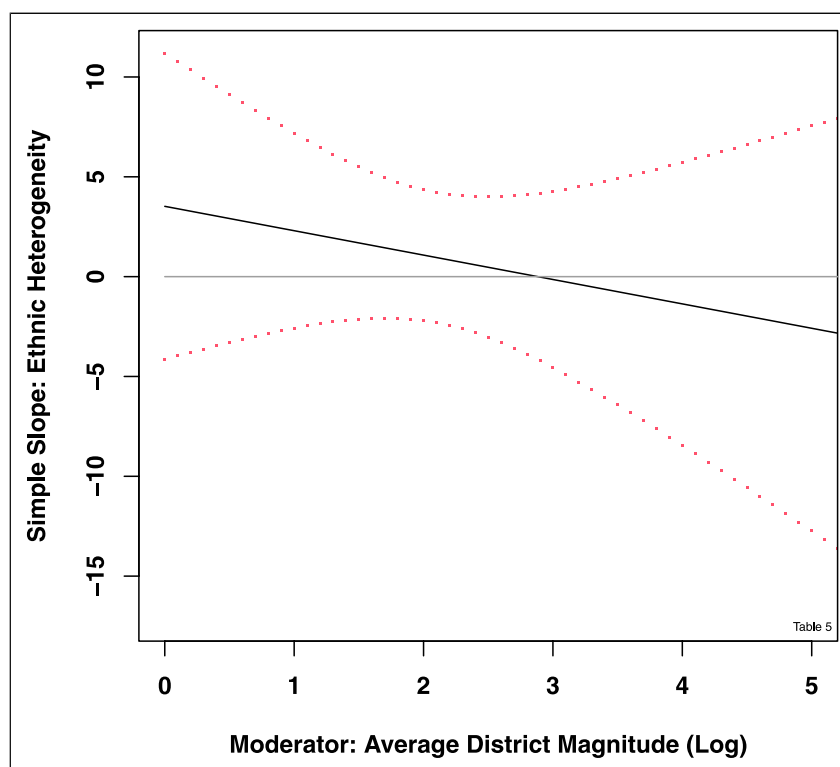


Figure 6. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude when Stability = 2. Note: Red dotted lines represent a 90% confidence interval.

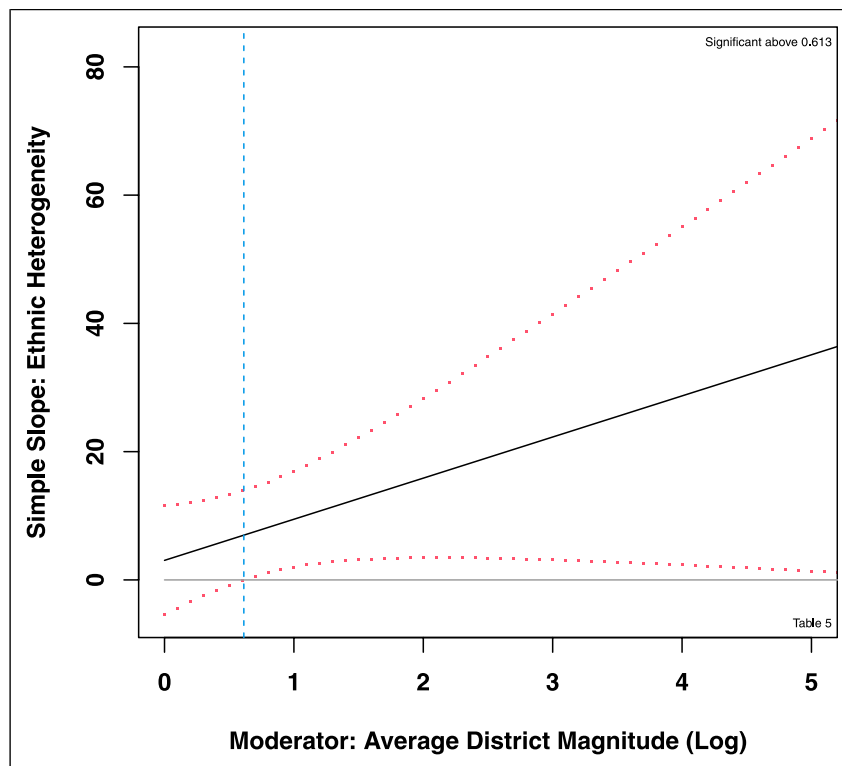


Figure 7. Marginal Effect of Ethnic Heterogeneity on the Effective Number of Electoral Parties by Average District Magnitude when Stability = 9. Note: Red dotted lines represent a 90% confidence interval.

consistent with the permissiveness of electoral rules and the level of social diversity.

Conclusions

We have explored the party system effects of electoral rules in highly unstable new democracies. We argued that even in shifting political environments the fragmentation of the party system is likely to become consistent with the permissiveness of electoral rules and the degree of social diversity when experience with elections increases and the rate of electoral regime change declines. These hypotheses were supported through the statistical analysis of 198 democratic elections in Latin America since the last inauguration of democracy up to 2021.

These findings do not imply that we can fully explain the dynamics of party system fragmentation in new democracies by simply considering the number of elections and electoral regime stability. A significant portion of the variation in party system fragmentation still remains unexplained. This is only natural, as many new democracies coexist with unstable social, political, and economic environments that upset

expected outcomes. This is apparent in Latin America, where election results are often affected by shifting political cleavages, economic crises, or corruption scandals. In such environments, politics and not just institutions shape electoral outcomes and constant variables, such as the index of ethno-linguistic fragmentation, are likely to be poor indicators of how changing social forces affect electoral preferences and put pressure on the emergence of new parties.

Nevertheless, it is a relevant contribution to show that even in highly volatile political environments new democracies eventually mature over time and party system fragmentation can be explained to a significant extent with the same variables used in established democracies. The two conditions for this development are electoral experience and electoral regime stability. While we confirm previous findings about the learning effects of successive elections, we show that these effects are stronger when elections take place under the same electoral rules. Electoral design matters amid changing social and political environments if politicians and voters are able and willing to let electoral institutions remain stable for a sufficiently long period of time.

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Data availability statement

Datasets and code for replication are available for download at: <https://doi.org/10.7910/DVN/QPA1T3> (Bunker and Negretto, 2023).

Supplemental Material

Supplemental material for this article is available online.

Notes

1. But see Li and Shugart (2016), and Shugart and Taagepera (2017), who argue that the predicted effects of electoral institutions should not differ significantly in new and old democracies.
2. In addition, there is no evidence of a meaningful interactive effect when a sample including established and new democracies from 1946 to 2000 is used.
3. Only an early study (Jones 1993) found some consistency between the expected outcomes of electoral formulas and district magnitude on party fragmentation.
4. The sample includes Argentina, Bolivia, Brazil, Chile, Colombia, Costa Rica, Dominican Republic, Ecuador, El Salvador, Guatemala, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay and Venezuela. Electoral data was collected based on official local sources.
5. Our coding follows the definition of democratic years of Boix et al. (2013).
6. In the absence of majoritarian electoral systems and with a few mixed electoral systems one cannot use an alternative specification focusing only on electoral reforms changing

between majoritarian and proportional or between types of mixed systems, as proposed by Lago and Martínez i Coma (2022).

7. Changes in the electoral cycle or in the presidential formula are observed indirectly, through shifts in the temporal proximity of elections and the effective number of presidential candidates.
8. In alternative tests (not shown) we considered the impact of the accumulated experience with elections in each country from 1900 to the year before the inauguration of democracy and found no significant effect, either when used as a control or as part of a new variable that adds past experience to elections after democratic transition. The most likely reason is that only a few countries (such as Chile or Uruguay) had several years of consecutive democratic elections before autocratic rule, which in turn interrupted electoral activity for a decade or more.
9. We obtained similar results considering the first half of the electoral cycle in each country as early elections or using a temporal division such as elections before or after 1999.
10. We found similar results using Shugart and Taagepera's formulas (derived from the seat product) for the predicted ENPV as regressors of the observed ENPV in simple and complex electoral systems. We reproduced the same analysis for the impact of electoral regime stability. See Tables A2 and A4 in the online appendix.

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